Lctr 18: System Analysis with the Laplace Transform and the Unilateral Laplace Transform

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Focus: Sections 9.7–9.9 (Pages 693–720)

18.1 The Big Picture

Why This Matters

In Lectures 16–17 we built the machinery: how to compute and invert Laplace transforms, and the properties that make them useful. Now we put it all to work. This lecture has two main goals:

Part I (System Analysis): Given a system described by a differential equation, find its transfer function $H(s)$, determine stability and causality from the pole locations and ROC, and compute the output for any input.

Part II (Unilateral Laplace Transform): Extend our tools to handle systems with **nonzero initial conditions**, the key to solving real-world transient problems where the system starts in some state other than rest.

By the end of this lecture, you will have a complete pipeline: differential equation → transfer function → pole-zero analysis → stability/causality determination → output computation, with or without initial conditions.

Roadmap:

1. The system function $H(s)$ for LTI systems described by differential equations.
2. Causality from the ROC.
3. Stability from the ROC (the $j\omega$-axis criterion).
4. Causality + stability together: the "all poles in the LHP" rule.
5. Block diagram representations.
6. The unilateral Laplace transform: definition and modified differentiation property.
7. Solving differential equations with initial conditions.

Part I: System Analysis with $H(s)$

18.2 The System Function $H(s)$

Consider a causal LTI system described by a linear constant-coefficient differential equation:

$$\sum_{k=0}^{N} a_k \frac{d^k y}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x}{dt^k}$$

Taking the bilateral Laplace transform of both sides (assuming zero initial conditions, i.e., the system is initially at rest):

$$\sum_{k=0}^{N} a_k \, s^k \, Y(s) = \sum_{k=0}^{M} b_k \, s^k \, X(s)$$

The **system function** (or **transfer function**) is:

$$\boxed{H(s) = \frac{Y(s)}{X(s)} = \frac{\sum_{k=0}^{M} b_k\, s^k}{\sum_{k=0}^{N} a_k\, s^k} = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_0}{a_N s^N + a_{N-1} s^{N-1} + \cdots + a_0}}$$

Key Insight

The Laplace transform converts the differential equation into an algebraic equation: each derivative d^k/dt^k becomes multiplication by s^k. The transfer function $H(s)$ is a **rational function**, a ratio of polynomials in s. The denominator roots are the poles; the numerator roots are the zeros. Everything about the system's behavior (stability, frequency response, transient response) is encoded in these pole and zero locations.

From Differential Equation to $H(s)$

System: $\dfrac{d^2y}{dt^2} + 5\dfrac{dy}{dt} + 6y = 2\dfrac{dx}{dt} + x$

Step 1: Replace $d^k/dt^k \to s^k$:

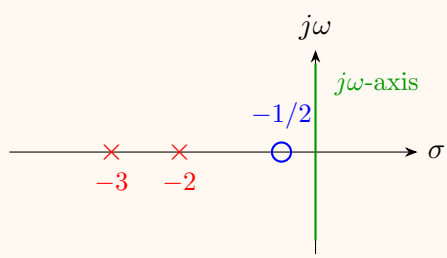
$$(s^2 + 5s + 6)\, Y(s) = (2s + 1)\, X(s)$$

Step 2: Form $H(s)$:

$$H(s) = \frac{2s+1}{s^2+5s+6} = \frac{2s+1}{(s+2)(s+3)}$$

Poles: $s = -2, -3$ (both in the LHP). Zero: $s = -1/2$.

Step 3: Pole-zero plot.



Both poles are in the LHP $\Rightarrow$ stable and causal (as we will formalize below).

18.3 Causality from the ROC

Causality Criterion

An LTI system is **causal** (output depends only on present and past input) if and only if:

1. The impulse response $h(t)$ is **right-sided**: $h(t) = 0$ for $t < 0$.

2. Equivalently, the ROC of $H(s)$ is a **right half-plane**:

$$\text{ROC}: \quad \text{Re}\{s\} > \sigma_{\max}$$

where $\sigma_{\max}$ is the real part of the rightmost pole.

3. If $H(s)$ is rational, causality also requires that $H(s) \to 0$ (or a finite constant) as $|s| \to \infty$, which means $M \leq N$ (degree of numerator $\leq$ degree of denominator).

18.4 BIBO Stability from the ROC

Stability Criterion

An LTI system is **BIBO stable** (bounded input $\Rightarrow$ bounded output) if and only if:

1. The impulse response is absolutely integrable: $\displaystyle\int_{-\infty}^{+\infty} |h(t)|\, dt < \infty$.

2. Equivalently, the ROC of $H(s)$ **includes the $j\omega$-axis**: $\text{Re}\{s\} = 0$ is in the ROC.

This means the Fourier transform $H(j\omega)$ exists, and the system has a well-defined frequency response.

18.5 Causal *and* Stable: The "All Poles in the LHP" Rule

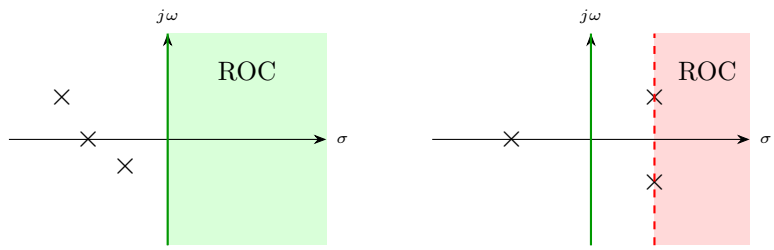
Combining the two criteria:

The Golden Rule for Causal LTI Systems

A causal LTI system with rational $H(s)$ is BIBO stable **if and only if all poles of $H(s)$ lie in the open left half-plane** (i.e., all poles have strictly negative real parts).

$$\text{Causal} + \text{Stable} \iff \text{Re}\{p_i\} < 0 \;\; \forall \text{ poles } p_i$$

Why this works: For a causal system, the ROC is $\text{Re}\{s\} > \sigma_{\max}$ (to the right of the rightmost pole). For this ROC to include the $j\omega$-axis, we need $\sigma_{\max} < 0$, which means every pole must have $\text{Re}\{p_i\} < 0$.



Causal + Stable
All poles in LHP
$j\omega$-axis in ROC ✓

Causal + Unstable
Some poles in RHP
$j\omega$-axis NOT in ROC ×

Stability Classification

Classify each system as stable, marginally stable, or unstable (assuming causality):

$H(s)$	**Poles**	**Pole locations**	**Classification**
$\frac{1}{s+3}$	$s=-3$	LHP	**Stable**
$\frac{1}{s-2}$	$s=2$	RHP	**Unstable**
$\frac{1}{s^2+4}$	$s=\pm j2$	$j\omega$-axis	**Marginally stable**
$\frac{s+1}{(s+2)(s+5)}$	$s=-2,-5$	LHP	**Stable**
$\frac{1}{(s+1)(s-1)}$	$s=\pm 1$	one in RHP	**Unstable**

Warning

Poles on the $j\omega$-axis (e.g., $s=\pm j\omega_0$) correspond to **marginally stable** systems: the impulse response neither decays nor grows but oscillates forever. These are on the boundary: technically not BIBO stable, because the $j\omega$-axis is *not* in the ROC (it is the ROC boundary). A sinusoidal input at the resonance frequency produces an unbounded output.

18.6 Complete System Analysis Pipeline

Complete Pipeline: Differential Equation to Output

System: $\frac{dy}{dt}+3y=x(t)$, **Input:** $x(t)=e^{-t}\,u(t)$.

Step 1: Find $H(s)$.

$$(s+3)\,Y(s)=X(s) \quad\Rightarrow\quad H(s)=\frac{1}{s+3}\,, \quad \text{ROC: } \mathrm{Re}\{s\}>-3$$

Pole at $s=-3$ (LHP) $\Rightarrow$ system is causal and stable. ✓

Step 2: Find $X(s)$.

$$X(s) = \frac{1}{s+1}, \quad \mathrm{Re}\{s\} > -1$$

Step 3: Find $Y(s) = H(s) \cdot X(s)$.

$$Y(s) = \frac{1}{(s+3)(s+1)}, \quad \mathrm{Re}\{s\} > -1$$

(The ROC is the intersection: $\sigma > -3 \cap \sigma > -1 = \sigma > -1$.)
Step 4: Partial fractions.

$$\frac{1}{(s+3)(s+1)} = \frac{A}{s+3} + \frac{B}{s+1}$$

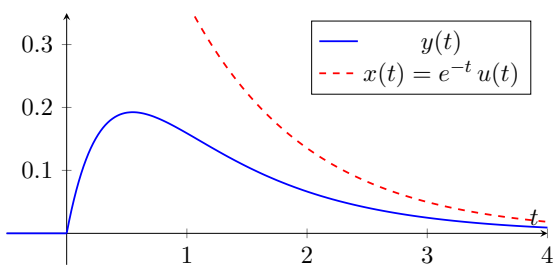
- $s = -3$: $1 = A(-3+1) \Rightarrow A = -1/2$.
- $s = -1$: $1 = B(-1+3) \Rightarrow B = 1/2$.

Step 5: Invert. Both poles are to the left of the ROC boundary ($\sigma = -1$), so both terms are right-sided:

$$\boxed{y(t) = \frac{1}{2}\left(e^{-t} - e^{-3t}\right) u(t)}$$

Step 6: Verify.

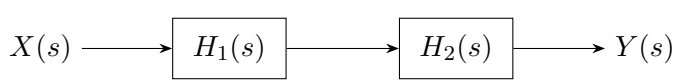
- $y(0^+) = \frac{1}{2}(1-1) = 0$ ✓ (system starts at rest)
- $y(\infty) = 0$ ✓ (both exponentials decay)
- $y'(0^+) = \frac{1}{2}(-1+3) = 1 = x(0^+)$ ✓ (from the ODE: $y'(0^+) + 3 \cdot 0 = 1$)



18.7 Block Diagram Representations

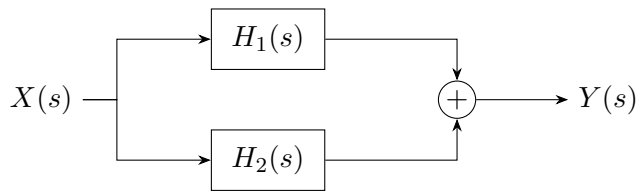
Complex systems are built from interconnections of simpler subsystems. In the s-domain:

18.7.1 Series (Cascade) Connection



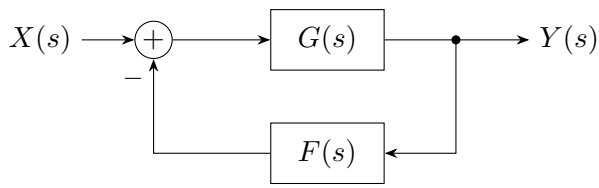
$$H(s) = H_1(s) \cdot H_2(s)$$

18.7.2 Parallel Connection



$$H(s) = H_1(s) + H_2(s)$$

18.7.3 Feedback Connection



$$H(s) = \frac{G(s)}{1 + G(s)\,F(s)}$$

Key Insight

Block diagrams in the s-domain are purely algebraic. Series connection = multiply. Parallel = add. Feedback = the formula above. This is far simpler than working with convolution integrals in the time domain, which is one of the main practical advantages of the Laplace transform.

Part II: The Unilateral Laplace Transform

18.8 Why We Need the Unilateral Transform

The bilateral Laplace transform assumes the system is **initially at rest** (all initial conditions are zero). But in practice, systems often have nonzero initial conditions:

- A capacitor charged to 5 V before a switch closes.
- A mass-spring system released from a displaced position.
- A control system with a nonzero set point at startup.

The **unilateral Laplace transform** handles these cases by modifying the lower limit of integration.

18.9 Definition

The Unilateral Laplace Transform

$$\boxed{\mathcal{X}(s) = \int_{0^-}^{\infty} x(t)\, e^{-st}\, dt}$$

The lower limit is 0^- (just before $t = 0$), which captures any impulse at $t = 0$. For signals that are zero for $t < 0$ (i.e., causal signals), the unilateral and bilateral transforms are identical.

Notation: We use $\mathcal{X}(s)$ or simply $X(s)$ for the unilateral transform when context is clear. Many textbooks use the same notation for both.

18.10 The Key Difference: The Differentiation Property

This is where the unilateral transform earns its keep. Recall that the bilateral transform gives:

$$\frac{dx}{dt} \;\mathcal{L}\; s\,X(s) \qquad \text{(bilateral, assumes zero ICs)}$$

The unilateral transform gives:

$$\boxed{\frac{dx}{dt} \;\mathcal{L}_u\; s\,\mathcal{X}(s) - x(0^-)}$$

For the second derivative:

$$\boxed{\frac{d^2x}{dt^2} \;\mathcal{L}_u\; s^2\,\mathcal{X}(s) - s\,x(0^-) - x'(0^-)}$$

General pattern for the n-th derivative:

$$\frac{d^n x}{dt^n} \;\; \mathcal{L}_u \;\; s^n\,\mathcal{X}(s) - s^{n-1}x(0^-) - s^{n-2}x'(0^-) - \cdots - x^{(n-1)}(0^-)$$

Key Insight

The initial conditions $x(0^-)$, $x'(0^-)$, etc. appear explicitly in the unilateral differentiation property. This is exactly the information needed to solve an ODE with specified initial conditions. Each derivative d^n/dt^n still becomes multiplication by s^n, but with correction terms that encode the initial state.

18.11 Solving Differential Equations with Initial Conditions

First-Order ODE with Nonzero Initial Condition

Solve: $\dfrac{dy}{dt} + 3y = 0\,, \quad y(0^-) = 5$

Step 1: Take the unilateral Laplace transform of both sides.

$$\big[s\,Y(s) - y(0^-)\big] + 3\,Y(s) = 0$$

$$s\,Y(s) - 5 + 3\,Y(s) = 0$$

Step 2: Solve for $Y(s)$.

$$(s+3)\,Y(s) = 5 \;\;\Rightarrow\;\; Y(s) = \frac{5}{s+3}$$

Step 3: Invert.

$$\boxed{y(t) = 5\,e^{-3t}\,u(t)}$$

The system decays from its initial value of 5 with time constant $\tau = 1/3$.

Check: $y(0) = 5$ ✓. $\quad y'(0) = -15$. From ODE: $y'(0) = -3y(0) = -15$ ✓.

Second-Order ODE with Initial Conditions

Solve: $\dfrac{d^2y}{dt^2} + 5\dfrac{dy}{dt} + 6y = 2\,u(t)\,, \quad y(0^-) = 1,\; y'(0^-) = 0$

Step 1: Unilateral Laplace transform.

For the left side:

$$\mathcal{L}_u\left\{\frac{d^2y}{dt^2}\right\} = s^2Y(s) - sy(0^-) - y'(0^-) = s^2Y - s - 0$$

$$\mathcal{L}_u\left\{5\frac{dy}{dt}\right\} = 5\big[sY(s) - y(0^-)\big] = 5sY - 5$$

$$\mathcal{L}_u\{6y\} = 6Y(s)$$

For the right side: $\mathcal{L}_u\{2\,u(t)\} = \dfrac{2}{s}$.

Step 2: Combine.

$$s^2Y - s + 5sY - 5 + 6Y = \frac{2}{s}$$

$$(s^2 + 5s + 6)\,Y(s) = \frac{2}{s} + s + 5$$

$$Y(s) = \frac{2/s + s + 5}{(s+2)(s+3)} = \frac{s^2 + 5s + 2}{s(s+2)(s+3)}$$

Step 3: Partial fractions.

$$\frac{s^2 + 5s + 2}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$

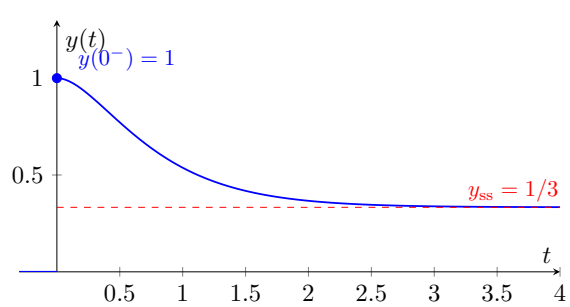
- $s = 0$: $2 = A(2)(3) \Rightarrow A = 1/3$.
- $s = -2$: $4 - 10 + 2 = B(-2)(1) \Rightarrow -4 = -2B \Rightarrow B = 2$.
- $s = -3$: $9 - 15 + 2 = C(-3)(-1) \Rightarrow -4 = 3C \Rightarrow C = -4/3$.

Step 4: Invert (all poles have ROC to the right, so all terms are right-sided):

$$\boxed{y(t) = \left[\frac{1}{3} + 2\,e^{-2t} - \frac{4}{3}\,e^{-3t}\right] u(t)}$$

Step 5: Verify.

- $y(0) = \frac{1}{3} + 2 - \frac{4}{3} = \frac{1+6-4}{3} = 1$ ✓
- $y'(0) = 0 + 2(-2) - \frac{4}{3}(-3) = -4 + 4 = 0$ ✓
- $y(\infty) = 1/3$ (the steady-state response to a step of amplitude 2, with DC gain $H(0) = 1/6$: $2 \times 1/6 = 1/3$) ✓



18.12 Zero-State Response + Zero-Input Response

The total response of a system with initial conditions can be decomposed:

$$\underbrace{y(t)}_{\text{total response}} = \underbrace{y_{\text{ZS}}(t)}_{\substack{\text{zero-state}\\ \text{response}}} + \underbrace{y_{\text{ZI}}(t)}_{\substack{\text{zero-input}\\ \text{response}}}$$

Zero-state response (ZSR): The output due to the input alone, assuming all initial conditions are zero. This is $Y_{\text{ZS}}(s) = H(s) \cdot X(s)$.

Zero-input response (ZIR): The output due to the initial conditions alone, with zero input. This is the "natural response" of the system from its initial state.

Key Insight

In the second-order example above, we can separate the two contributions:
ZSR (set $y(0^-) = 0$, $y'(0^-) = 0$, keep input $2u(t)$):

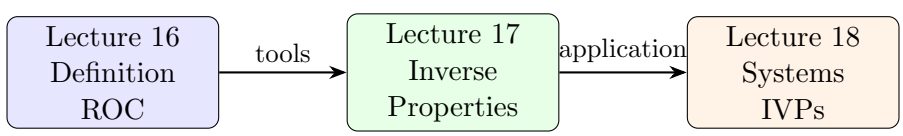
$$Y_{\text{ZS}}(s) = \frac{2/s}{(s+2)(s+3)} = \frac{2}{s(s+2)(s+3)}$$

ZIR (set input to 0, keep $y(0^-) = 1$, $y'(0^-) = 0$):

$$Y_{\text{ZI}}(s) = \frac{s+5}{(s+2)(s+3)}$$

Adding: $Y_{\text{ZS}} + Y_{\text{ZI}} = \frac{s^2+5s+2}{s(s+2)(s+3)}$, which is exactly what we computed. The decomposition helps you understand *which part of the output comes from the input and which comes from the initial conditions.*

18.13 Summary of the Three-Lecture Block



1. $H(s)$ **from a differential equation:** Replace d^k/dt^k with s^k and rearrange. $H(s) = Y(s)/X(s)$ is a rational function whose poles and zeros fully characterize the system.
2. **Causality:** The system is causal iff the ROC is a right half-plane (to the right of the rightmost pole) and $M \leq N$.
3. **Stability:** The system is BIBO stable iff the ROC includes the $j\omega$-axis. For a causal system, this reduces to: **all poles in the open LHP**.
4. **Block diagrams:** Series = multiply, parallel = add, feedback = $G/(1+GF)$.
5. **Unilateral Laplace transform:** The differentiation property $dx/dt \leftrightarrow sX(s) - x(0^-)$ incorporates initial conditions. This converts an IVP (differential equation + initial conditions) into an algebraic equation.
6. **Total response = ZSR + ZIR:** Zero-state (due to input) plus zero-input (due to ICs).

18.14 Common Mistakes to Avoid

1. **Confusing bilateral and unilateral transforms:** The bilateral transform assumes zero ICs. If the problem gives nonzero ICs, you *must* use the unilateral transform (or account for ICs explicitly).

2. **Forgetting IC terms in the unilateral differentiation property:** $\mathcal{L}_u\{y'\} = sY - y(0^-)$, **not** just sY. For second derivatives, there are *two* IC terms.

3. **Checking stability from the wrong criterion:** For a causal system, stability $\Leftrightarrow$ all poles in LHP. For a non-causal system, this rule does not apply; you must check whether the $j\omega$-axis is in the (possibly non-right-half-plane) ROC.

4. **Concluding instability from RHP zeros:** Zeros in the RHP do *not* cause instability. Only **poles** in the RHP (for causal systems) cause instability. RHP zeros affect the transient response shape (nonminimum-phase behavior) but not stability.

5. **Applying the pipeline to noncausal systems without adjusting the ROC:** If the system is not causal, the ROC is not necessarily a right half-plane. Always determine the ROC from the signal type (right-sided, left-sided, or two-sided).

6. **Mixing up $y(0^-)$ and $y(0^+)$:** The unilateral Laplace transform uses 0^- (just *before* $t = 0$) as the initial condition. If the input has a discontinuity at $t = 0$ (like a step), $y(0^+)$ may differ from $y(0^-)$.

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